7 solution	276	1	1	$\Delta H = -3268.12 + \left(6 - \frac{7}{2}\right) X\, 8.314\, X\, 10^{-3}\, X\, 298$ $\Delta U = -3268.12 + 2.5\, X\, 8.314\, X\, 10^{-3}\, X\, 298$ $\Delta U = -3268.12 + 6.19$ $\Delta U = -3261.93\ kJ\ mol^{-1}$	$\Delta H = -3268.12 + \left(6 - \frac{15}{2}\right) X\, 8.314\, X\, 10^{-3}\, X\, 298$ $\Delta H = -3268.12 - 1.5\, X\, 8.314\, X\, 10^{-3}\, X\, 298$ $\Delta H = -3268.12 - 3.72$ $\Delta H = -3271.84\ kJ\ mol^{-1}$
7	281	-	Last two lines	In $CaCO_3$ (s) $\longrightarrow$ CaO(s) + CO_2(g) , entropy change is positive, In (a) and (b) entropy change is negative ; in (c) entropy change is zero.	In option (d) ,A solid gives a gaseous product , hence entropy change is expected to be maximum.
7	287	1	Q.NO 67	T = 33K $N_2O_4 \rightleftharpoons 2NO_2$ Initial concentration 100% 0 Concentration dissociated 50% - Concentration remaining at equilibrium 50% 100% $K_{eq} = \frac{100}{50} = 2$ $\Delta G^0 = -2.303\ RT \log K_{eq}$ $\Delta G^0 = -2.303 \times 8.314 \times 33 \times \log 2$ $\Delta G^0 = -190.18\ J\ mol^{-1}$	<table><tr><td></td><td>N_2O_4</td><td>⇌</td><td>$2NO_2$</td></tr><tr><td>Initial no of moles</td><td>100</td><td></td><td>0</td></tr><tr><td>Number of moles dissociated</td><td>50</td><td></td><td>-</td></tr><tr><td>No of moles remaining</td><td>50</td><td></td><td>100</td></tr><tr><td>Total no of moles</td><td colspan="3">150 moles</td></tr><tr><td>$P_{N_2O_4} = \frac{n_{N_2O_4}}{n_{N_2O_4} + n_{NO_2}} . P$ $P_{N_2O_4} = \frac{50\ mol}{150\ mol} .1\ atm$ $P_{N_2O_4} = 0.333\ atm$ $P_{NO_2} = \frac{n_{NO_2}}{n_{N_2O_4} + n_{NO_2}} . P$ $P_{NO_2} = \frac{100\ mol}{150\ mol} .1\ atm$ $P_{NO_2} = 0.667\ atm$</td><td colspan="3">$K_P = \frac{P_{NO_2}^2}{P_{N_2O_4}}$ $K_P = \frac{(0.667)^2\ atm^2}{0.333\ \text{atm}}$ $K_P = 1.336\ atm$ $\Delta G^0 = -2.303\ RT \log K_P$ ΔG^0 $= -2.303\ X\ 8.314\ X\ 33\ X\ \log 1.336$ $\Delta G^0 = -79.49\ J\ mol^{-1}$</td></tr></table>

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